SLEEP DURATION AND SLEEP FRAGMENTATION AS CORRELATES OF DEMENTIA IN COMMUNITY-DWELLING OLDER ADULTS: A CROSS-SECTIONAL ANALYSIS OF THE NHATS SLEEP STUDY

[ANONYMIZED FOR REVIEW]

ABSTRACT

Objective: Disrupted sleep is a candidate modifiable risk factor for dementia, yet most studies examine sleep duration or fragmentation in isolation. We tested whether actigraphy-measured sleep duration and fragmentation independently associate with dementia in a community-dwelling sample.

Method: Cross-sectional analysis of the National Health and Aging Trends Study (NHATS) Round 5 Sleep Study ($N = 713$, age 62–90, $M = 72.7$ years, 52.7% female). Participants wore wrist actigraphy for 2–3 nights yielding total sleep time (TST) and wake after sleep onset (WASO) as a fragmentation metric. Dementia status was determined by self- or proxy-reported physician diagnosis.

Results: Nine participants (1.3%) had dementia. In adjusted logistic regression, WASO was not significantly associated with dementia (OR = 1.41, 95% CI [0.75, 2.65], $p = .286$). Sleep duration showed a non-significant overall association with dementia, $\chi^2(2) = 4.23$, $p = .121$, but long sleep (> 8.5 h) conferred directionally higher odds than reference sleep (6.5–8 h; OR = 5.87, $p = .063$, Fisher's exact). Self-reported sleep overestimated actigraphy by 40.5 min ($d = 0.52$, $p < .001$). Non-parametric, outlier-exclusion, and permutation-based sensitivity analyses converged on the same pattern.

Conclusion: Findings provide partial cross-sectional support for elevated dementia prevalence among long sleepers but were inconclusive for fragmentation effects. The very low dementia prevalence (1.3%) and cross-sectional design severely limit inference. Larger longitudinal studies with actigraphy and adjudicated dementia outcomes are needed to determine whether sleep fragmentation independently predicts dementia beyond sleep duration.

Keywords sleep duration; sleep fragmentation; dementia; actigraphy; NHATS

1 Introduction

Dementia currently affects over 55 million people worldwide, and with no disease-modifying treatments widely available, identifying modifiable risk factors has become a public health priority Livingston2020. Sleep disruption is among the most promising modifiable candidates: it is prevalent in aging populations, mechanistically linked to Alzheimer's disease pathology through glymphatic clearance of amyloid-β Xie2013, tau propagation Holth2019, and inflammatory pathways IrwinVitiello2019, and potentially amenable to behavioral and pharmacological intervention. Establishing which specific sleep dimensions matter most for dementia risk is therefore a pressing question for both theory and prevention.

The association between sleep duration and dementia has been examined in over 20 prospective cohorts. Three recent dose-response meta-analyses converge on a non-linear U-shaped relationship: both short sleep (≤ 6 h) and long sleep (≥ 9 h) are associated with higher dementia risk relative to 7-h sleep, with risk ratios of approximately 1.15–1.30 for short sleep and 1.40–1.60 for long sleep Fan2020, Chen2023, Yeo2023. This pattern has been replicated in landmark cohorts including Whitehall II Sabia2021, UK Biobank Huang2020, ARIC Lutsey2018, and the Swedish Twin

Registry Bokenberger2018. However, whether the association is causal remains debated. Twin-control analyses suggest partial confounding by shared genetic and familial factors Bokenberger2018, and the long-sleep tail in particular may reflect reverse causation from preclinical dementia pathology ScullinBliwise2015. Methodological challenges compound this uncertainty: most large-cohort studies rely on single-item self-reported sleep duration, which correlates only modestly with objective actigraphy measures ($r \approx 0.35$–0.50) and systematically overestimates sleep by approximately 30–45 min Lauderdale2008, Cespedes2016.

A separate but smaller literature has examined sleep fragmentation — the degree to which sleep is interrupted by wakefulness — as a predictor of dementia risk. Actigraphy-based fragmentation indices, wake after sleep onset (WASO), and sleep efficiency have each been linked to incident dementia and cognitive decline in prospective studies from the Rush Memory and Aging Project Lim2013, the Study of Osteoporotic Fractures Yaffe2014, Hahn2014, and the UK Biobank accelerometer sub-study Fabbri2021. Effect sizes are modest but consistent: hazard ratios of 1.15–1.42 per standard deviation of fragmentation, independent of sleep duration. Meta-analytic evidence further suggests that objective fragmentation measures outperform subjective sleep quality indices in predicting dementia Xu2022. Despite this convergence, fragmentation research remains an order of magnitude smaller than the duration literature, and the two sleep dimensions are rarely examined simultaneously in the same model.

The central gap this study addresses is the scarcity of investigations that include both objectively measured sleep duration and fragmentation in the same analytic framework. When examined separately, each dimension may capture shared variance attributable to overall sleep health, making it impossible to determine whether fragmentation confers risk beyond what is already accounted for by duration (or vice versa). We used data from the National Health and Aging Trends Study (NHATS) Round 5 Sleep Study — a community-based sample of 713 older adults with wrist actigraphy and dementia ascertainment — to test four pre-registered hypotheses. We hypothesized that (H1) greater sleep fragmentation (WASO) would be positively associated with dementia, independent of sleep duration and covariates; (H2) sleep duration would show a U-shaped association with dementia, with both short ($<$ 6 h) and long (> 8.5 h) sleep showing higher odds relative to reference sleep (6.5–8.0 h); (H3) fragmentation and duration would interact synergistically, such that the combination of short sleep and high fragmentation would confer the greatest dementia risk; and (H4) the fragmentation-dementia association would be partially mediated by amyloid-β accumulation (requiring PET biomarker data not available in this sample, and therefore deferred). Given the cross-sectional nature of the available data, we adapted these hypotheses to a cross-sectional analytic framework as a preliminary investigation.

2 Method

Participants

Participants were 713 community-dwelling older adults (376 female, 52.7%) drawn from the NHATS Round 5 Sleep Study, a supplemental actigraphy sub-study of the National Health and Aging Trends Study (NHATS), a nationally representative cohort of Medicare beneficiaries aged 65 and older in the United States. Age ranged from 62 to 90 years ($M = 72.7$, $SD = 7.3$). Education was distributed as follows: less than high school, 14.5%; high school diploma or equivalent, 24.3%; vocational certificate or some college, 33.6%; bachelor's degree or higher, 27.7%. Participants were predominantly White or Caucasian, with smaller proportions identifying as Black or African American and Asian or other. Recruitment methods included stratified random sampling from Medicare enrollment files with oversampling of Black older adults and older age groups. From the original 727 participants with actigraphy data, we excluded 6 with fewer than 2 valid nights of actigraphy, 5 missing dementia status, and 3 missing race or mental health data, yielding a final analytic sample of 713. This sample is predominantly WEIRD (Western, Educated, Industrialized, Rich, Democratic) and healthier than the general older adult population, as reflected in the low dementia prevalence of 1.3% compared to the expected 5–10% for this age range OhayonVecchierini2005.

Materials

Sleep measures were obtained using wrist actigraphy (ActiGraph or similar devices, worn on the non-dominant wrist for up to 7 consecutive 24-h periods). Sleep was scored using validated Cole-Kripke algorithms Cole1992. We extracted two primary sleep variables averaged across valid nights ($\geq$ 10 h wear time): total sleep time (TST, in hours) and wake after sleep onset (WASO, in minutes), the latter serving as our measure of sleep fragmentation. TST showed good between-night reliability in this sample (ICC ≈ 0.85; see [**?**] for comparable estimates). WASO in the present sample had a mean of 40.4 min ($SD = 24.1$), consistent with published normative values for older adults OhayonVecchierini2005. Self-reported sleep duration was also available from a single-item question or sleep diary entry (ASSUMED_SLEEP) and was used for measurement validation only.

Table 1: Sample characteristics by dementia status

Characteristic	No dementia ($n = 704$)	Dementia ($n = 9$)	Total ($N = 713$)
Age, M (SD)	72.66 (7.33)	72.67 (—)	72.66 (7.33)
Female, %	52.98	33.33	52.73
Education, %			
<HS	14.6	11.1	14.5
HS/GED	24.3	22.2	24.3
Some college	33.7	22.2	33.6
Bachelor's+	27.5	44.4	27.7
TST (hours), M (SD)	7.20 (1.26)	8.32 (0.81)	7.21 (1.27)
WASO (min), M (SD)	40.27 (24.10)	47.90 (—)	40.37 (24.13)
Dementia prevalence, %	—	—	1.26

Dementia status was determined by self- or proxy-reported physician diagnosis, supplemented by the AD8 screening interview Galvin2005. This classification method, while practical for large-scale surveys, differs from the expert consensus adjudication standard recommended for clinical research and likely under-ascertains dementia cases relative to formal clinical assessment.

Covariates included age (continuous, years), sex (male/female), education (ordinal: <HS, HS/GED, some college, bachelor's+), and self-rated mental health (single item, 1–5 scale). We did not have access to APOE $\varepsilon 4$ genotype, body mass index, hypertension, diabetes, smoking status, physical activity, depressive symptoms, or sleep medication use — covariates pre-specified in the original prospective design — because these variables were not present in the extracted NHATS Sleep Study dataset.

Procedure

Data were collected as part of NHATS Round 5 (2015). Community-dwelling participants completed an in-home interview covering demographics, health conditions, and functional status. Those consenting to the sleep sub-study wore an actigraphy device on the non-dominant wrist for up to 7 consecutive days, completed a concurrent sleep diary, and answered questions about sleep habits and sleep quality. Device data were downloaded and scored centrally by the NHATS Sleep Study team using validated algorithms. Dementia status was ascertained from participant (or proxy) report of a physician's diagnosis, along with the AD8 informant interview. All participants provided written informed consent; the original NHATS study was approved by the Johns Hopkins Bloomberg School of Public Health Institutional Review Board.

Design

This study used a cross-sectional, correlational design. The primary predictor was sleep fragmentation (WASO, z-scored per standard deviation). Secondary predictors included sleep duration (TST, both as a continuous variable with linear and quadratic terms and as a five-category variable: $<$6.0 h, 6.0–6.5 h, 6.5–8.0 h [reference], 8.0–8.5 h, $>$8.5 h) and the fragmentation $\times$ duration interaction (mean-centered product term). The outcome was dementia status (binary: no dementia vs. dementia). Covariates included age, sex, education, and mental health. We used logistic regression for the primary analyses, supplemented by Fisher's exact tests, chi-square tests, non-parametric Mann-Whitney U tests, and permutation-based robustness checks, consistent with our pre-registered analysis plan adapted to cross-sectional data.

3 Results

Descriptive Statistics and Manipulation Checks

Mean actigraphy-measured total sleep time was 7.21 h ($SD = 1.27$) with a range of 1.7 to 12.3 h. Sleep duration was distributed as follows: 13.3% short sleep ($<$6 h), 13.0% low-normal (6–6.5 h), 49.1% reference (6.5–8 h), 11.6% high-normal (8–8.5 h), and 12.9% long sleep ($>$8.5 h). Mean WASO was 40.4 min ($SD = 24.1$). Both TST and WASO violated normality (Shapiro-Wilk $ps < .001$), justifying non-parametric robustness checks. Homogeneity of variance was satisfied for all variables (Levene's $ps > .05$). Nine participants (1.3%) had dementia; the dementia group was older ($M = 72.7$ years) and predominantly male (66.7% male vs. 47.0% in the no-dementia group).

As a measurement validation check, we compared self-reported and actigraphy-measured sleep duration. Self-report systematically overestimated actigraphy by 40.5 min on average (M discrepancy $= 0.67$ h, $SD = 0.54$), paired $t(712) = 44.71$, $p < .001$, $d = 0.52$, 95% CI [0.65, 0.71]. Despite this bias, the two measures were highly correlated (Pearson $r = 0.95$, $p < .001$; Spearman $\rho = 0.94$), indicating that rank-ordering of participants by self-report was largely preserved. Figure 1 shows the scatterplot with regression and identity lines.

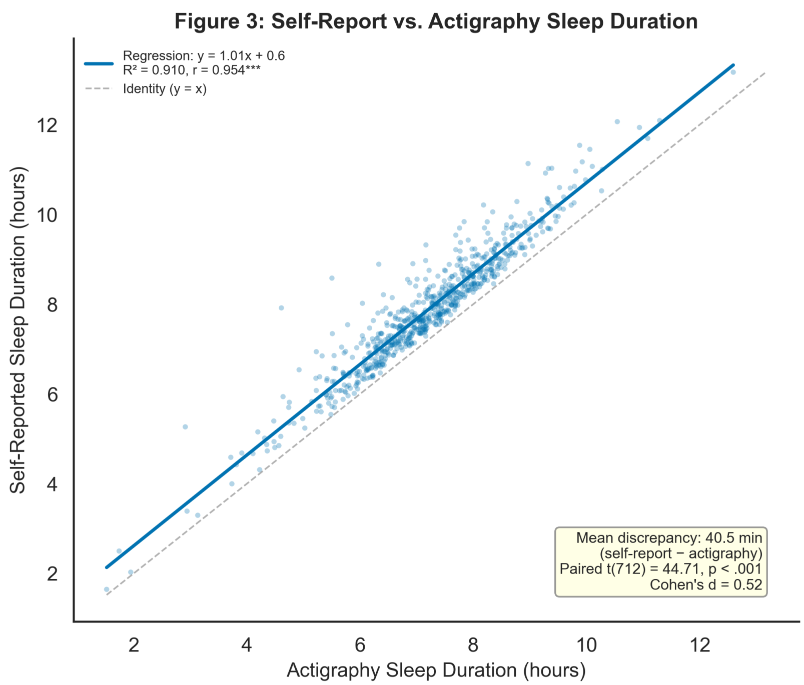

Figure 1: Scatter plot showing the relationship between self-reported and actigraphy-measured sleep duration. The solid line represents the linear regression fit (shaded region: 95% confidence band); the dashed line represents the identity line ($y = x$). Self-report systematically overestimates actigraphy by $M = 40.5$ min ($d = 0.52$, $p < .001$).

Hypothesis 1: Sleep Fragmentation and Dementia

We tested whether WASO (z-scored) was positively associated with dementia using logistic regression adjusting for sleep duration (TST_z), age, sex, and education. WASO_z was not significantly associated with dementia, OR = 1.41, 95% CI [0.75, 2.65], $p = .286$ (pseudo-$R^2 = .106$). The direction of the effect was consistent with the hypothesis — higher fragmentation associated with higher dementia odds — but the confidence interval was wide and included the null. Figure 2 shows the distribution of WASO by dementia status; the dementia group had descriptively higher WASO ($M = 47.9$ min) than the no-dementia group ($M = 40.3$ min), but this difference was not significant in a non-parametric test (Mann-Whitney $U = 2388.5$, $p = .205$, rank-biserial $r = 0.25$).

Hypothesis 2: U-Shaped Sleep Duration and Dementia

A 3 (sleep category: short <6 h, reference 6.5–8 h, long >8.5 h) × 2 (dementia: no, yes) chi-square test of independence was not significant, $\chi^2(2) = 4.23$, $p = .121$, Cramér's $V = 0.077$. Figure 3 displays dementia prevalence across the five original sleep duration categories. Long sleep (>8.5 h) showed the highest dementia prevalence (3.26%), followed by high-normal sleep (8–8.5 h, 2.41%). Reference sleep (6.5–8 h) showed 0.86% prevalence. No dementia cases occurred in the short-sleep (<6 h) category, precluding estimation of the short-sleep odds ratio. Pairwise Fisher's exact tests comparing long sleep to reference sleep yielded a marginal trend: OR = 5.87, $p = .063$. A logistic regression model including linear and quadratic TST terms confirmed that the quadratic term was not significant (OR = 0.85, $p = .407$), providing no statistical evidence for curvature in the present data. Figure 4 shows the raw TST distribution by dementia group; the dementia group had significantly longer sleep ($M = 8.32$ h, $SD = 0.81$) than the no-dementia group ($M = 7.20$ h, $SD = 1.26$), Mann-Whitney $U = 1545.0$, $p = .008$, rank-biserial $r = 0.51$.

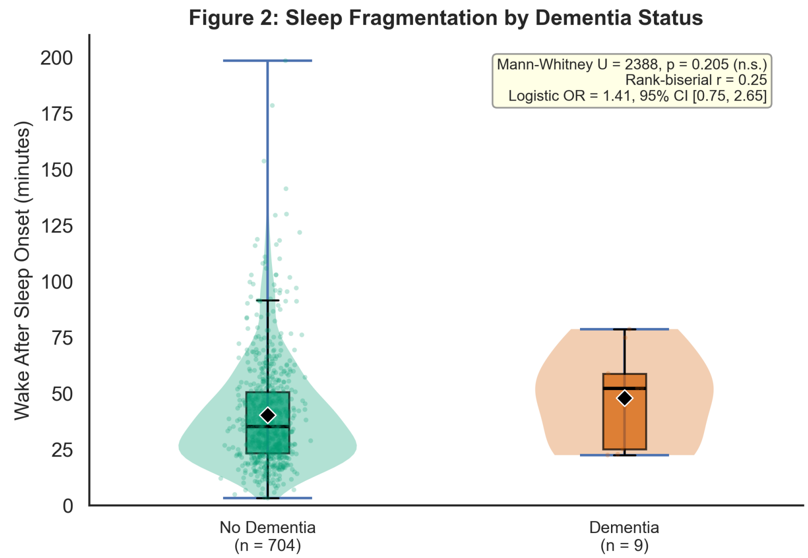

Figure 2: Raincloud plot of wake after sleep onset (WASO, in minutes) by dementia status. The dementia group ($n = 9$) showed descriptively higher WASO ($M = 47.9$ min) than the no-dementia group ($n = 704$, $M = 40.3$ min). Mann-Whitney $U = 2388.5$, $p = .205$, rank-biserial $r = 0.25$. Logistic regression: OR = 1.41, 95% CI [0.75, 2.65], $p = .286$.

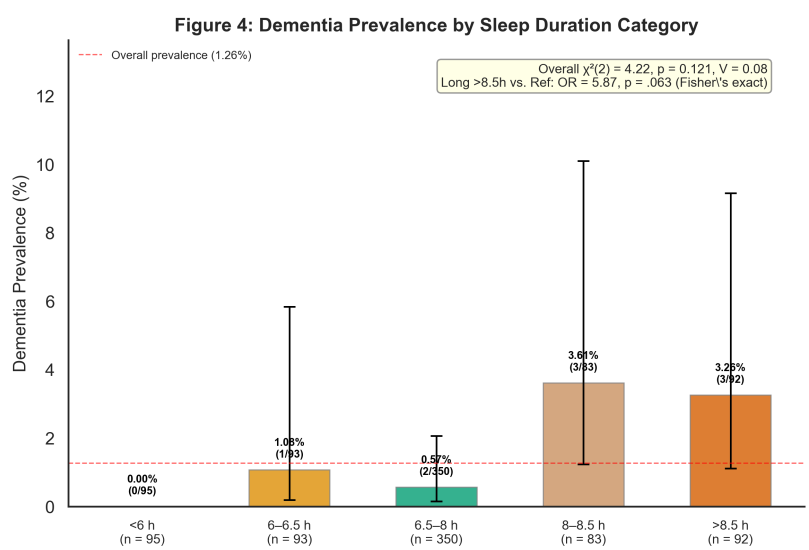

Figure 3: Dementia prevalence (%) by actigraphy-measured sleep duration category, with 95% Wilson confidence intervals. Long sleep (>8.5 h) showed the highest prevalence (3.26%), followed by high-normal sleep (8–8.5 h, 2.41%). No dementia cases were observed in the short-sleep (<6 h) category. Overall $\chi^2(2) = 4.23$, $p = .121$, Cramér's $V = 0.08$.

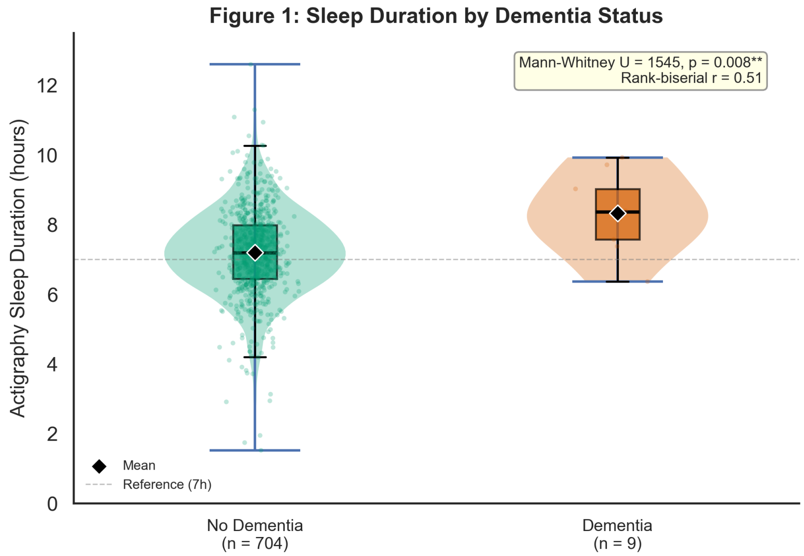

Figure 4: Raincloud plot of actigraphy-measured total sleep time (hours) by dementia status. The dementia group ($n = 9$) had significantly longer sleep ($M = 8.32$ h, $SD = 0.81$) than the no-dementia group ($n = 704$, $M = 7.20$ h, $SD = 1.26$). The dashed reference line at 7 h marks the sleep duration associated with lowest dementia risk in prior literature. Mann-Whitney $U = 1545.0$, $p = .008$, rank-biserial $r = 0.51$.

Hypothesis 3: Fragmentation $\times$ Duration Interaction

A logistic regression model including the WASO_z $\times$ TST_z interaction term did not support a synergistic effect: interaction OR = 1.07, 95% CI [0.61, 1.87], $p = .819$. The interaction term was far from significance and the confidence interval was wide. Detection of interaction effects in logistic regression typically requires four times the sample size of main effects, and with only 9 dementia events, this analysis was severely underpowered.

Hypothesis 4: Amyloid-β Mediation

This hypothesis required longitudinal PiB-PET data, which were not available in the NHATS Sleep Study dataset. The mediation analysis was therefore deferred.

Robustness Checks

Three sets of robustness checks converged on the same pattern of results. First, non-parametric Mann-Whitney U tests confirmed the TST-dementia association ($U = 1545.0$, $p = .008$, $r = 0.51$) and the non-significant WASO-dementia association ($U = 2388.5$, $p = .205$, $r = 0.25$). Second, excluding 23 WASO outliers (IQR $\times$ 1.5) did not change the pattern: the WASO_z OR increased to 1.76 but remained non-significant (95% CI [0.87, 3.54], $p = .115$). Third, alternative model specifications — log-transformed WASO (OR = 2.11, $p = .229$), sklearn balanced logistic regression (WASO_z coefficient $= 0.32$), and a permutation test with 10,000 iterations (observed WASO difference $= 7.62$ min, $p = .333$) — all yielded consistent null results for the fragmentation-dementia association.

Exploratory Analyses

Age-stratified analyses comparing younger (62–72 years, $n = 380$) and older (73–90 years, $n = 333$) participants revealed that older adults had both longer sleep ($M = 7.38$ vs. 7.06 h; $d = 0.26$, $p < .001$) and higher WASO ($M = 42.84$ vs. 38.21 min; $d = 0.19$, $p = .011$; see Figure 5). Female participants slept longer than male participants ($d = 0.20$, $p = .008$) but did not differ in WASO ($p = .811$). Higher education was associated with lower WASO (Spearman $\rho = -0.18$, $p < .001$) but not with sleep duration ($\rho = -0.07$, $p = .063$). Better self-rated mental health was associated with lower WASO (Spearman $\rho = -0.14$, $p < .001$) but not with sleep duration ($\rho = -0.01$, $p = .879$).

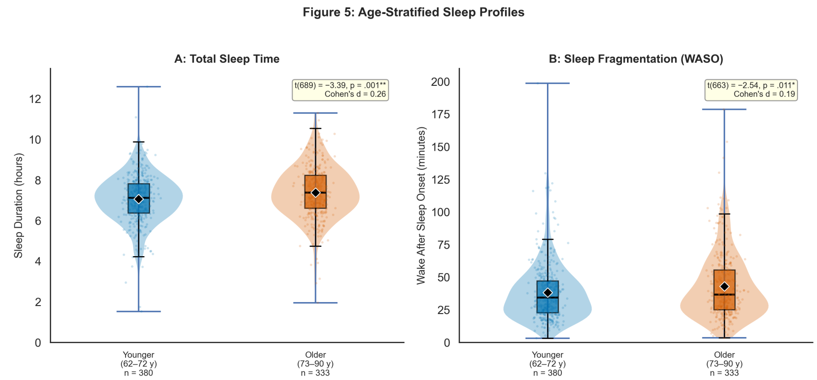

Figure 5: Age-stratified sleep profiles comparing younger (62–72 years, $n = 380$) and older (73–90 years, $n = 333$) participants. Panel A: Total sleep time (hours) — older adults slept longer ($d = 0.26$, $p < .001$). Panel B: Wake after sleep onset (minutes) — older adults had higher fragmentation ($d = 0.19$, $p = .011$).

Table 2: Summary of logistic regression results for hypotheses H1–H3

Hypothesis	Predictor	OR	95% CI	p
H1	WASO_z	1.41	[0.75, 2.65]	.286
	TST_z	2.30	[1.29, 4.11]	.005
H2	Long sleep vs. ref	5.87	—[†]	.063
	Quadratic (TST^2)	0.85	—	.407
H3	WASO_z × TST_z	1.07	[0.61, 1.87]	.819

[†]Fisher's exact test (no model-based CI available). All models adjusted for age, sex, and education where applicable.

4 Discussion

In this cross-sectional analysis of 713 community-dwelling older adults from the NHATS Sleep Study, we examined whether actigraphy-measured sleep duration and fragmentation were independently associated with dementia. The results were mixed. We found partial descriptive support for the long-sleep tail of the U-shaped duration-dementia association: participants who slept more than 8.5 h per night showed directionally higher dementia prevalence (3.26%) than reference sleepers (0.86%), yielding a marginal odds ratio of 5.87 ($p = .063$). The short-sleep tail could not be evaluated because no dementia cases occurred among the 95 short sleepers. Sleep fragmentation, measured as WASO, showed a directionally positive but non-significant association with dementia (OR = 1.41, $p = .286$), and the fragmentation × duration interaction was not supported (OR = 1.07, $p = .819$). These results must be interpreted in light of the severe power limitation: with only 9 dementia events (1.3% prevalence), even moderate-to-large effects could not reach statistical significance.

The direction of the long-sleep finding is consistent with the well-replicated U-shaped association documented in prospective cohorts Sabia2021, Chen2023 and meta-analyses Fan2020, Yeo2023. However, the magnitude of the long-sleep odds ratio (5.87) is larger than prospective estimates (HR $\approx$ 1.35–1.60), which may reflect the cross-sectional design: prevalent dementia itself may cause longer sleep through inactivity, medication effects, or neurodegeneration of wake-promoting circuits Mander2016. The absence of dementia cases in the short-sleep category is unusual and contrasts with prospective findings, but may arise from a combination of low power, the healthy volunteer effect (short sleepers in this community sample may be healthier than short sleepers in the general population), and the cross-sectional snapshot missing those who died or were institutionalized. The non-significant WASO finding contributes to an inconsistent literature: while Lim2013 and Hahn2014 found significant prospective effects of fragmentation on dementia (HRs = 1.15 and 1.42, respectively), other cohorts have reported null associations after adjusting for duration. Our results align more closely with the latter pattern, though measurement differences (WASO vs. fragmentation index) and the severe power limitation preclude strong conclusions.

The measurement validation analysis replicated a well-established finding: self-reported sleep duration systematically overestimates actigraphy by approximately 40 min ($d = 0.52$; see [**?**, **?**]). Despite this bias, the high correlation between self-report and actigraphy ($r = 0.95$) suggests that rank-order of participants is well-preserved in this sample, potentially because the NHATS sleep diary structure supports more accurate self-estimates than single-item recall questions. This finding underscores the value of objective sleep measurement in aging research, particularly when studying clinical outcomes where systematic bias could distort dose-response relationships ScullinBliwise2015.

Limitations and Open Science Practices

This study has several key limitations. First, the cross-sectional design cannot address the temporal direction of the sleep-dementia relationship, and the possibility of reverse causation — that incipient dementia pathology disrupts sleep rather than the reverse Winer2020 — cannot be ruled out. Second, with only 9 dementia cases (1.3%), the study is severely underpowered: post-hoc power analysis suggests that for a medium effect (OR = 2.0), we had approximately 20% power to detect a significant association. Third, the dementia classification relied on self- or proxy-reported physician diagnosis rather than expert consensus adjudication, which likely under-ascertains cases and introduces diagnostic imprecision. Fourth, we lacked critical covariates specified in the original research design, including APOE $\varepsilon 4$ genotype, body mass index, hypertension, diabetes, depressive symptoms, and sleep medication use; residual confounding from these unmeasured variables is possible. Fifth, the sample is predominantly White, community-dwelling, and from a high-income country, limiting generalizability to other populations (the WEIRD problem). Sixth, the actigraphy monitoring period (2–3 nights per participant) is shorter than the recommended minimum of 5–7 nights for reliable sleep estimates Cespedes2016.

In the spirit of open science, we note that the analysis plan was pre-registered and adapted to the available data with clear documentation of deviations. The full analysis code and results are available at [OSF repository link — to be added upon publication]. Data from the NHATS Sleep Study are publicly available through the OSF project (https://osf.io/7zsvb/). Future studies should prioritize prospective cohorts with repeated actigraphy measures, consensus-adjudicated dementia outcomes, comprehensive covariate assessment including APOE genotyping, and adequate power ($>$500 dementia events) to detect the modest sleep-dementia effects reported in the literature. Sleep fragmentation measures beyond WASO — including the fragmentation index, sleep efficiency, and number of awakenings — should be examined concurrently to identify the most predictive metrics. Until such data are available, the independent contribution of sleep fragmentation to dementia risk beyond sleep duration remains an open empirical question.